Factors

1 State whether the following are factors of $33wy$:

a 11 **b** $66wy$ **c** yw **d** $3w$

e yw^2 **f** 33

2 Find the highest common factor of the following groups of numbers:

a 2 and $6x$ **b** 3 and $15x$

c $6x$ and 21 **d** 10 and $16u$

e $35x$ and 14 **f** $4x$ and $16xy$

g 15 and $5x$ **h** $24m$, $12n$ and $18q$

i $27m, 24n$ and $15q$

Factorisation

3 Complete the factorisation of the following expressions:

a $2a + 16 = \square\,(a + 8)$ **b** $5y - 30 = \square\,(y - 6)$

c $24n + 15 = \square\,(8n + 5)$ **d** $16a - 6 = \square\,(8a - 3)$

e $6v + 18 = 6\,(\square)$ **f** $9a - 36 = 9\,(\square)$

g $24a + 9 = 3\,(\square)$ **h** $6w - 27 = 3\,(\square)$

i $4r + 12 = \square\,(r + 3)$ **j** $5v - 30 = \square\,(v - 6)$

k $18c - 24 = \square\,(3c - 4)$ **l** $2r + 14 = 2\,(\square)$

m $5t - 10 = 5\,(\square)$ **n** $54u + 12 = 6\,(\square)$

4 Factorise the following expressions:

a $6v + 30$ **b** $6a - 48$ **c** $45t - 40$ **d** $2m + 10n$

e $54 - 60fg$ **f** $8g - 72h$ **g** $9f + 15g$ **h** $3w + 15$

i $2c - 18$ **j** $40c - 35$ **k** $6w + 30y$ **l** $14y + 4w$

m $4a - 14b$ **n** $32 - 36jk$ **o** $8jkt - 18$ **p** $3fg - 10fh$

q $3rtw - 8rw$ **r** $4rt - 6r^2t$ **s** $16wy - 36w^2y$

5 Complete the following factorisation:

$$2w - 10z + 8y = 2\,(\square - \square + \square)$$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 Factorise the following completely:

a $8g + 28h - 20$

b $4w + 4z + 4y$

c $4u - 12v + 20w$

d $10m - 25n - 15p$

e $3m - 6n + 12p$

f $10g - 35h + 15$

g $zw - zy - zk$

h $60r - 25rt - 35r^2t$

i $12m - 32mn - 20m^2n$

j $16z - 32yz - 24y^2z$

k $25r^2t - 15rt^2 - 45r^2t^2$

Factorisation of negatives

7 Factor out the negative coefficient of the following expressions:

a $-4r - 3s$

b $-7r - 7s$

c $-2 - 9r$

d $-8x - 5y$

8 Factorise the following expressions:

a $-8b - 16$

b $-18a + 16$

c $-80v - 30$

d $-36 - 6x$

e $-25u - 35v$

f $-35m + 49n$

g $-8pqr - 6pr$

h $-8w^2 + 3w^2y$

i $-24j^2 - 27j^2k$

j $-8m^2 + 9mn$

k $-10u^2v + 9uv^2$

l $-21j^2 - 36j^2k$

m $-2s - 10$

n $-8v^2 + 56$

o $-12s + 10$

p $-28b - 21$

q $-63r^2 + 28$

r $-4m - 32n$

s $-49y - 21w$

9 Factorise the following completely:

a $-18p - 9q - 45r$

b $-10r - 25s + 20t$

c $-20u - 25v + 45w$

d $-8w - 18y - 10z$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647